Software Development and Professional Practice

Process Life Cycle Models

By

Dr. Rasha Mahmoud Kamel

Scrum

- The idea in the agile Scrum methodology is that a ***small team is unified around a single goal*** and gets together for sprints of development that move them towards that goal.
- Like other agile methodologies, Scrum emphasizes the efficiency of small teams (***no more than 10 developers***), direct communication, iterative development, learning from frequent feedback, and collective ownership.
- **Scrum** is ***a project management methodology***. It ***doesn’t define many of the detailed development practices*** (like pair programming or test-driven development) that XP does. Despite this, Scrum teams typically use many of the XP practices. Common code ownership, pair programming, small releases, simple design, test-driven development, continuous integration, and coding standards are all common practices in Scrum projects.

Scrum Sprints

- Scrum is characterized by the **sprint** which is ***an iteration with a predetermined duration*** of commonly 1–4 weeks.
- Sprints are ***time-boxed*** in that they are of a ***fixed duration*** and the output of a sprint is what work the team can accomplish during the sprint.
- The delivery date for the sprint ***does not move out***. The scope of work to be delivered needs to be chosen so as to accommodate this delivery date.
- This means that sometimes a sprint can be terminated early if all work planned for the sprint has been completed, and sometimes a sprint will finish with less functionality than was proposed. A sprint always delivers ***a usable product***.

Scrum Artifacts

- Scrum requirements are encapsulated in two backlogs:
  - The **Product Backlog** is the prioritized list of all the requirements for the project. The product owner creates and prioritizes the product backlog composed of user stories, based on a desired Product Goal. The development team breaks the high-priority user stories into tasks and estimates them. This list of tasks becomes the Sprint Backlog.
  - The **Sprint Backlog** is the prioritized list of user stories for the current sprint. Once the sprint starts, ***only the development team can add tasks to the sprint backlog***; these are usually bugs found during testing. No outside entity can add items to the sprint backlog, only to the Product Backlog. The Sprint Goal is a guiding star for the team to follow. Thus, any changes to the Sprint Backlog during the sprint should support this goal.

Scrum Roles

- Scrum explicitly defines three roles in a development project:

  (1) Product Owner.

  (2) Development Team.

  (3) Scrum Master.

Product Owner

- The Product Owner is responsible for ***defining and prioritizing the product requirements.***
- These requirements are typically ***expressed as user stories*** and are maintained in the Product Backlog.
- Each story can be summarized into sentences like “As a <type of user>, I would like to <do something>, so that <I achieve a goal>.” For example, “As a student, I would like to be able to submit my assignments online so that I can meet deadlines without having to come to the office”.
- The Product Owner ***orders the backlog items according to business value*** and ensures that the Development Team clearly understands the requirements behind each user story.

Scrum Master

- The Scrum Master is an ***expert in Scrum*** who facilitates Scrum projects.
- The Scrum master ***helps*** the team understand and apply Scrum values, principles, and practices.
- The Scrum master ***protects*** the team from external interruptions during the sprint.
- The Scrum master ***ensures*** that everyone makes progress by removing impediments that hinder the team's progress.
- The Scrum master ***ensures*** that the Daily Scrum meetings and other Scrum events are held. The Scrum Master ***helps*** keep Daily Scrum meetings short and focused. The Scrum Master ***records*** the decisions made at the meeting and tracks action items.

Development Team

- The Development Team consists of all ***professionals doing the work to deliver a potentially releasable increment***, such as developers, designers, testers, and so on.
- The team reviews the user stories generated and prioritized by the Product Owner, and breaks them into tasks to plan how the features will be implemented. The team selects a subset of stories and tasks that can be completed within the given sprint duration.
- The team ***is self-organizing***; the developers ***decide among themselves who will work on which user stories and tasks***, and what ***the development process*** they will employ during the sprint.
- The team collectively owns all code and the project and shares the goal of delivering a potentially releasable product increment at the end of each sprint.

Scrum Meetings

- Scrum prescribes different types of meetings:

  (1) Sprint planning meeting.

  (2) Daily Scrum meeting.

  (3) Sprint retrospective meeting.

  (4) Backlog refinement (grooming) meeting.

Sprint Planning Meeting

- The sprint planning meeting is ***held at the beginning of each sprint*** to define the work that will be performed during the upcoming sprint. **During this meeting:**
  - The Product Owner presents ***the highest-priority items*** from the Product Backlog.
  - The developers ***estimate the Product Backlog items***, typically using story points.
  - Then, the developers review these items and ***select the ones they believe they can complete during the sprint***, based on their **historical velocity** (the total story points completed in previous sprints) and their **current capacity**.
  - The selected items ***move from the Product Backlog to the Sprint Backlog***.
  - The team then ***breaks these items into tasks and estimates those tasks in hours*** to ensure that the total planned work does not exceed the team's available capacity.
  - A ***Sprint Goal is established*** to provide a shared objective and focus for the sprint.

Sprint Planning Meeting

- It is generally recommended that tasks in a sprint be small enough (***no longer than one day of effort***) to improve tracking and transparency. If a task is estimated to take more than one day of effort, it is often divided into two or more tasks.
- The sprint begins immediately after Sprint Planning. During the sprint, the team works toward achieving the Sprint Goal and completing the Sprint Backlog items.
- **After each sprint,** a new Sprint Planning Meeting is held where the Product Owner and the team ***re-prioritize the product backlog*** and ***create a backlog for the next sprint***.
- This planning meeting is also where the organization can decide whether ***the project is finished***, or whether to finish the project at all.

Daily Scrum Meeting

- The Daily Scrum is held every day of the Sprint often at the beginning of the day. It is ***a stand-up meeting of*** **15–30** ***minutes*** duration and is conducted by the developers.
- During this meeting, each team member talks about:
  - What tasks have been finished yesterday.
  - What is planned to be done today.
  - What impediments blocking their progress.
- This meeting:
  - Allows the team to ***share information*** and ***track sprint progress***.
  - ***Reinforces team spirit*** by sharing progress.
  - ***Highlights problems*** so that any delays in the schedule or any implementation issues are immediately obvious and can be addressed by the team at once.

Sprint Retrospective Meeting

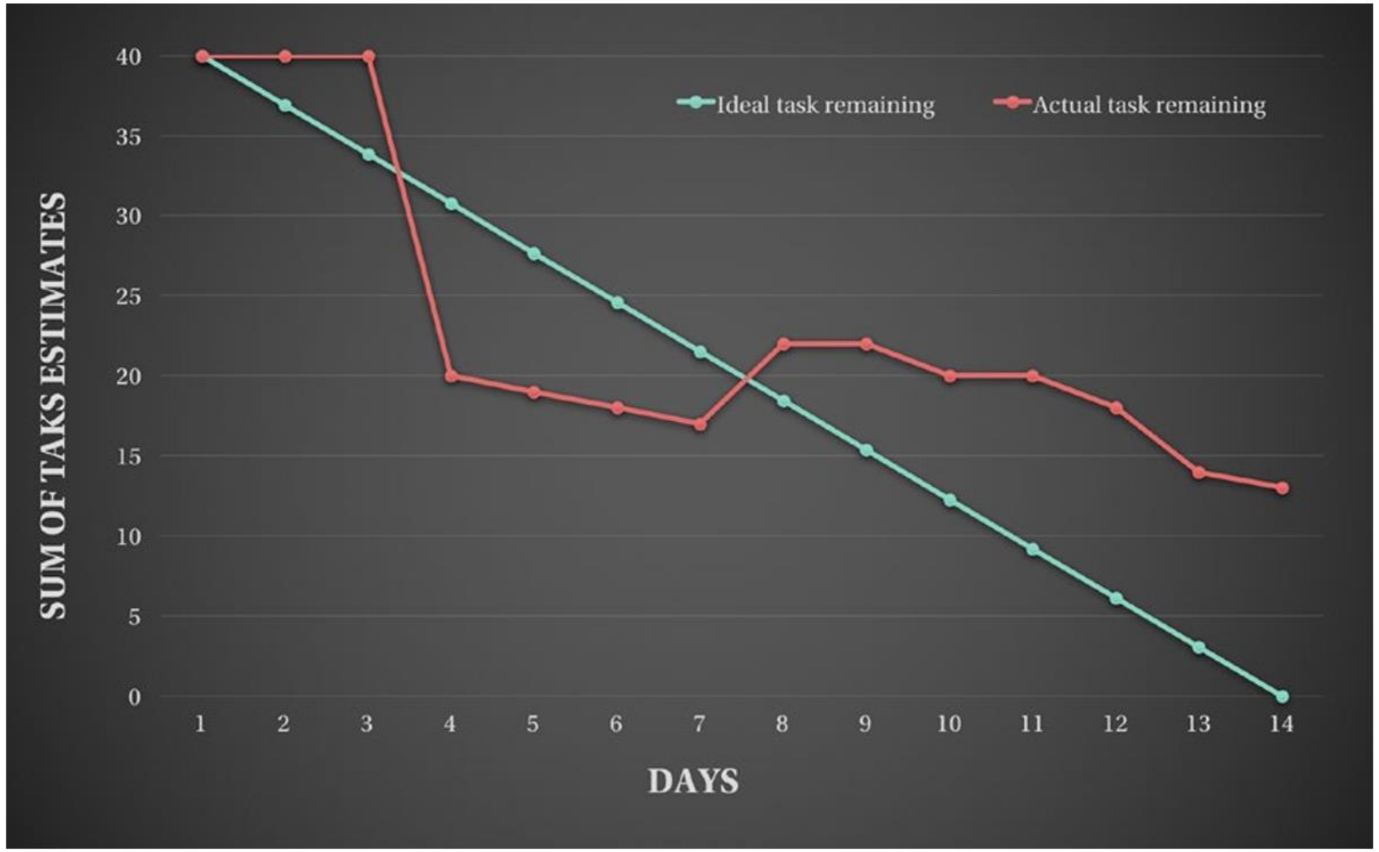
- Once the sprint is over, the ***current version of the product is released to the Product Owner,*** who may perform acceptance testing on it.
- The team holds a sprint retrospective meeting to ***ponder the just completed sprint and analyze whether or not it has been successful*** and understand what made it like that.
- During this meeting, the team reflects on the completed sprint. They discuss
  - What went well during the sprint.
  - What problems were encountered during the sprint.
  - What can be done to avoid these problems and improve performance in the future sprints.
- During this meeting, the Scrum Master ***calculates the team's velocity*** and ***analyzes the burndown chart.***

Sprint Retrospective Meeting

- The retrospective meeting should ***end with several action points***.
- These points must be visible to everyone, have a responsible person assigned, and ideally a deadline.
- During the next retrospective, the team will analyze the progress on the action points from the previous retrospective meeting.

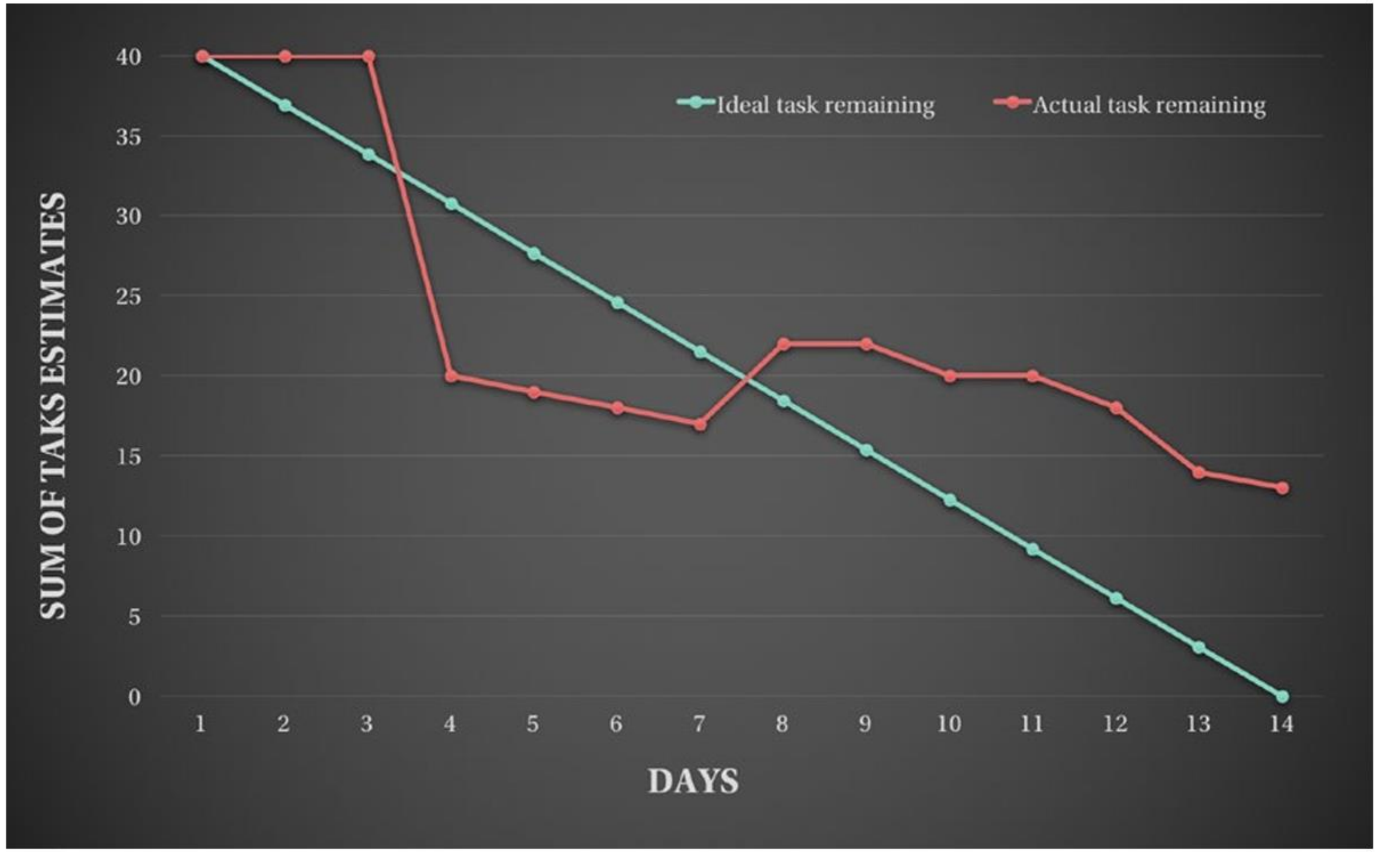
Burndown Chart

- The burndown chart is a chart that establishes the relationship between the ***amount of committed*** and ***amount of completed work***.
- The x-axis shows the number of days in a sprint—in this example, 14.
- The y-axis represents the number of estimated tasks—in this example, we assume that tasks have been estimated in story points and that there are 40 story points of tasks in the sprint.



Burndown Chart

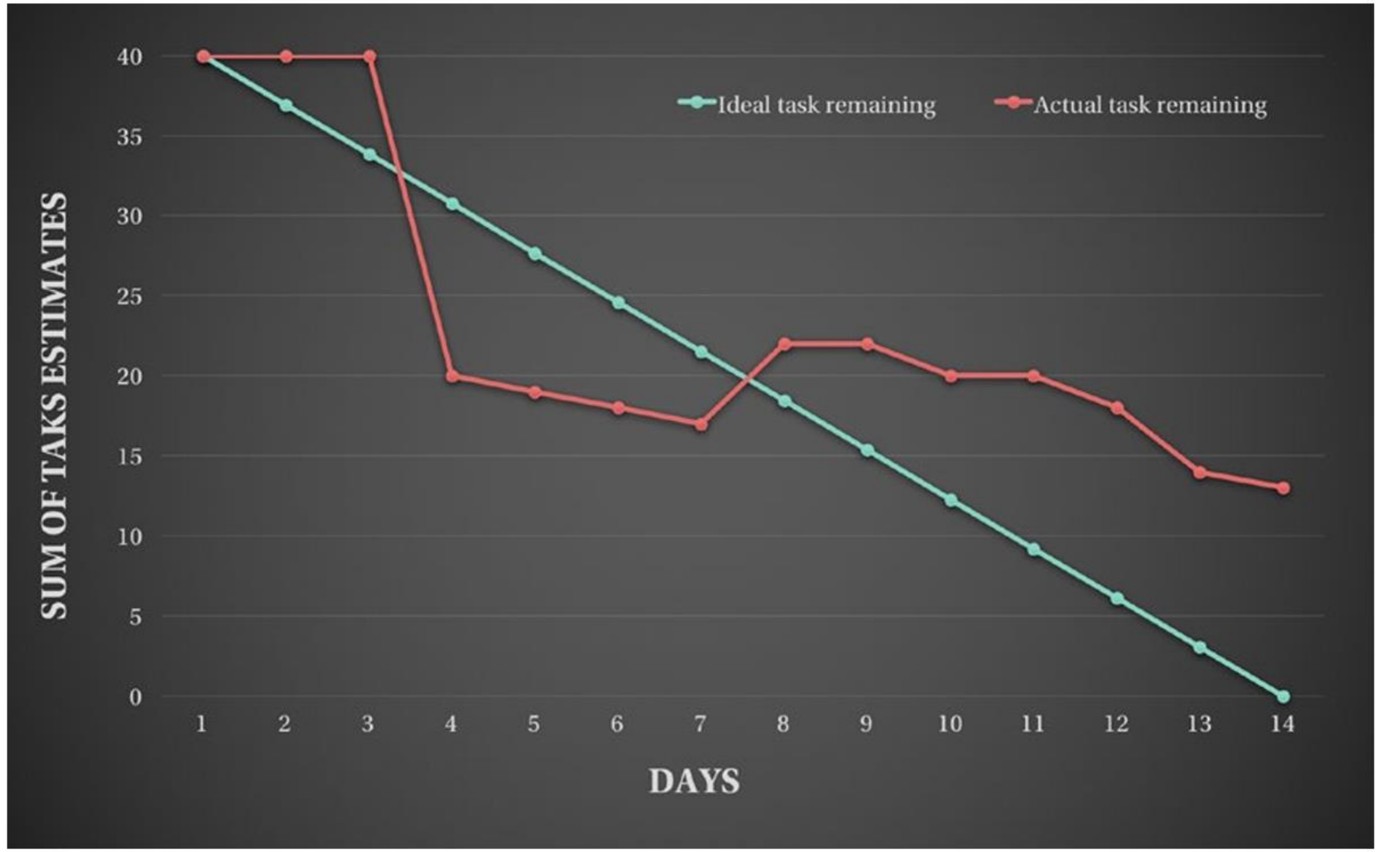
- The **green line** represents the scrum Nirvana; all the tasks that have been planned and estimated in the sprint planning have been finished until the end of the sprint with a well-balanced uniform pace.
- The **red line** represents the actual state of the sprint. Analyzing the chart in the end of the sprint gives an idea of how the sprint went.



- The more the red line is closer to the green, the more ideal the sprint is.

Burndown Chart

- In this example, we can guess that for the first 2 days nothing really happened, and suddenly the team went really productive on the third day, then there has been a constant pace during the next 3 days, then someone committed a crime and added some tasks to the sprint on the eighth day, and after that there was almost a constant pace until the end of the sprint but still some tasks remained unfinished.

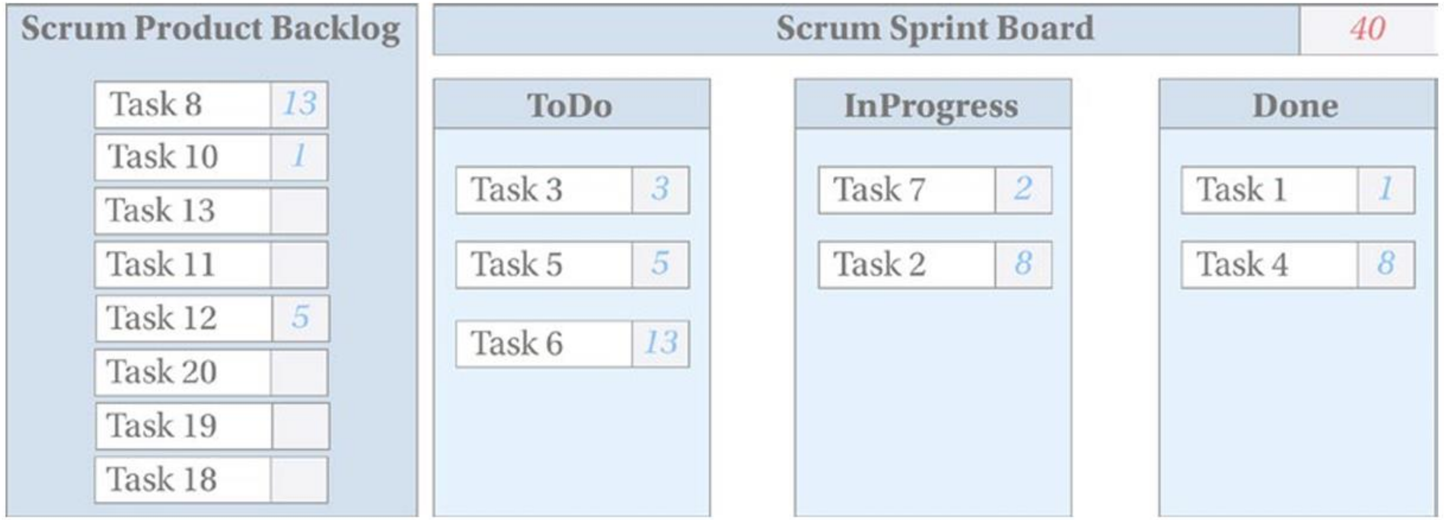


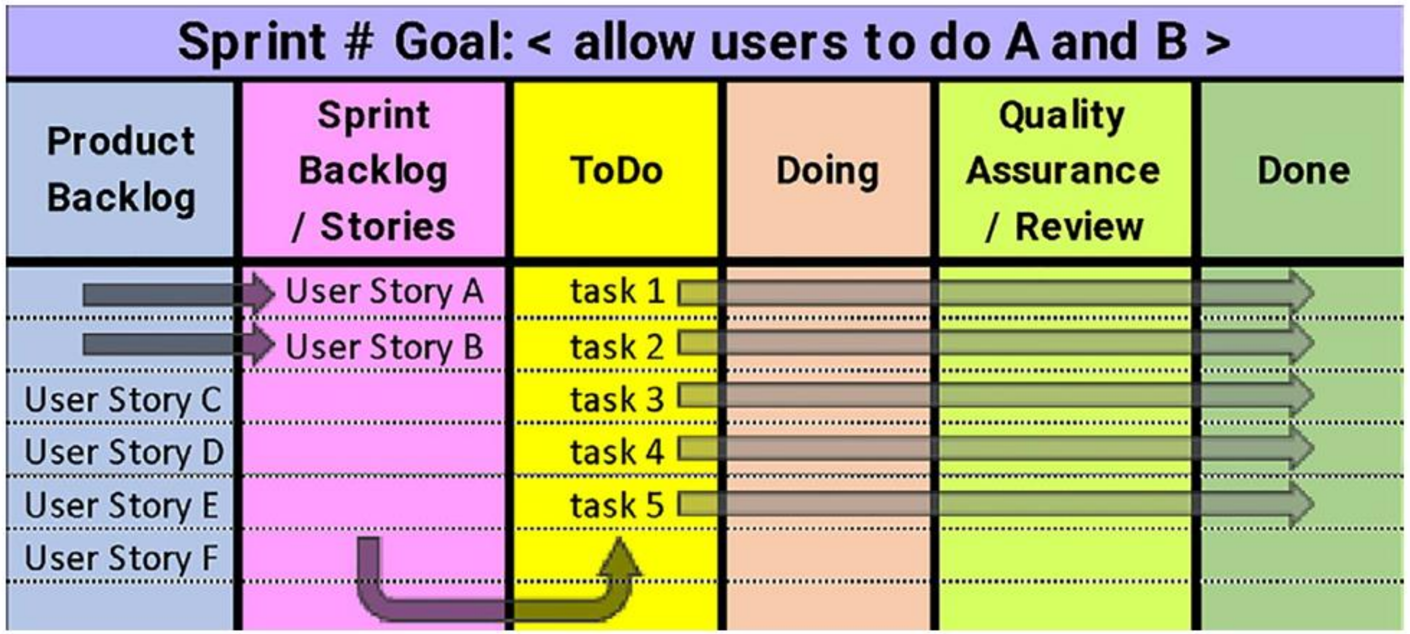
Backlog Refinement Meeting

- The backlog refinement (grooming) meeting is held ***periodically throughout the sprint*** (such as once a week) to ***refine backlog items*** and make the backlog clean, ensuring they are ready for future sprint planning.
- During this meeting:
  - The Product Owner can add items to the backlog and re-prioritize some items and move them to the top of the backlog.
  - The team ensures that the stories are clear and the acceptance criteria is achievable.
  - Developers breaks down high priority stories into tasks.
  - Each task is carefully analyzed and redefined if needed.
  - The tasks can also be broken into several subtasks, depending on the size and the needs.

Scrum Board

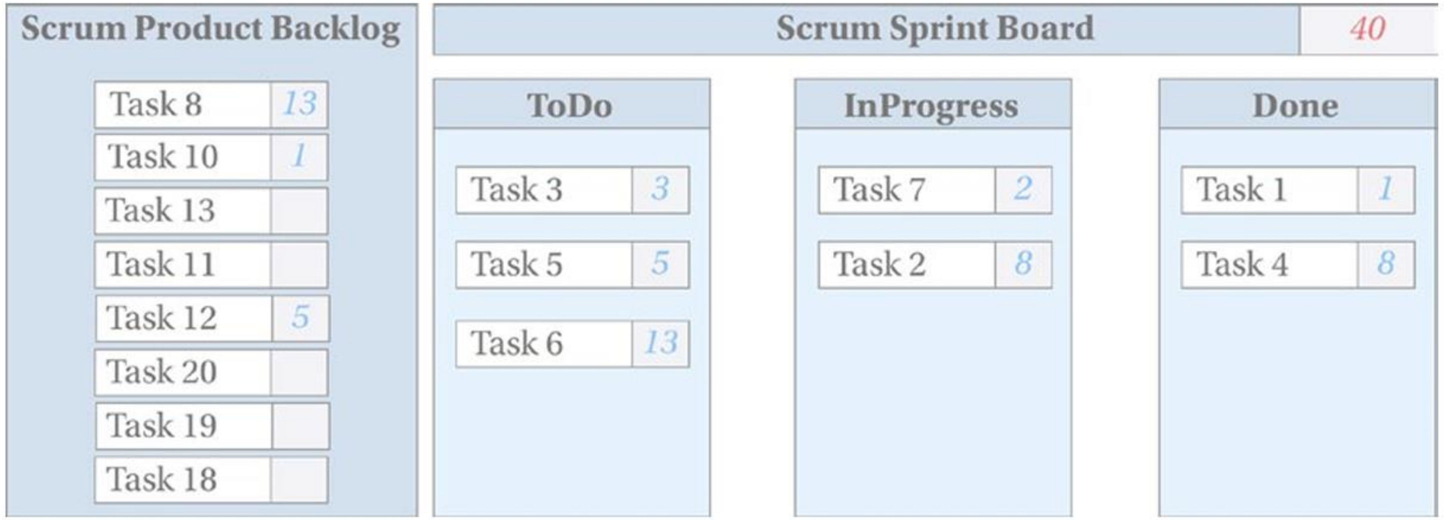
- The ***Sprint Backlog is usually visual*** and provides everyone with a clear shared view of the progress status of the sprint work.
- It is commonly represented on a task board using Post-It notes or index cards, with one card per task. This board includes at least three columns:
  - ToDo
  - InProgress
  - Done

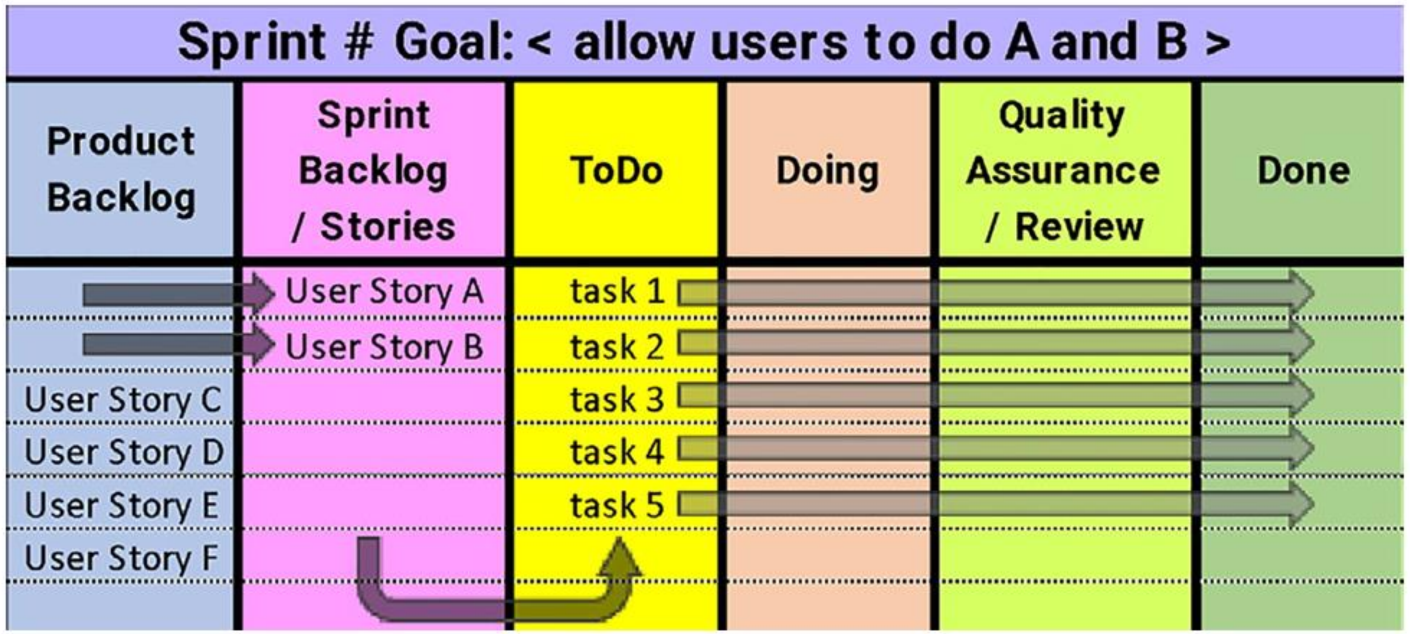




Scrum Board

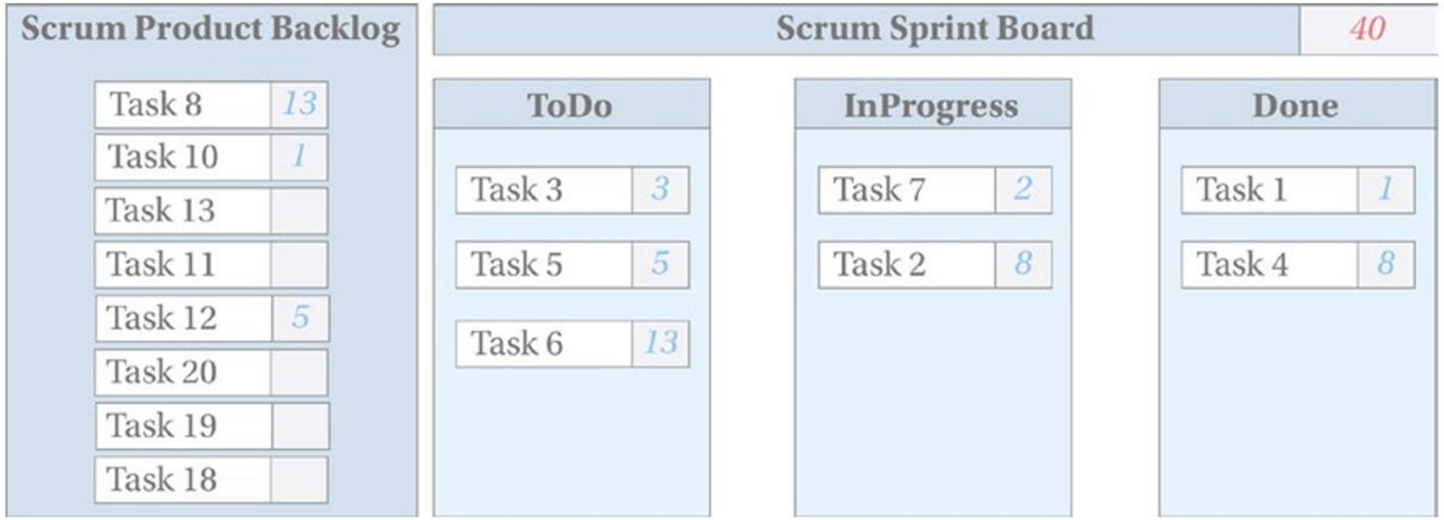
- During the sprint, the tasks move from the **ToDo** state to **Done** when it satisfies the team’s definition of done.
- Some teams consider the task done when the code is complete, while others might consider the task done once the tests are passed.
- However, it makes sense to consider the task done once it is already deployed and running in production.

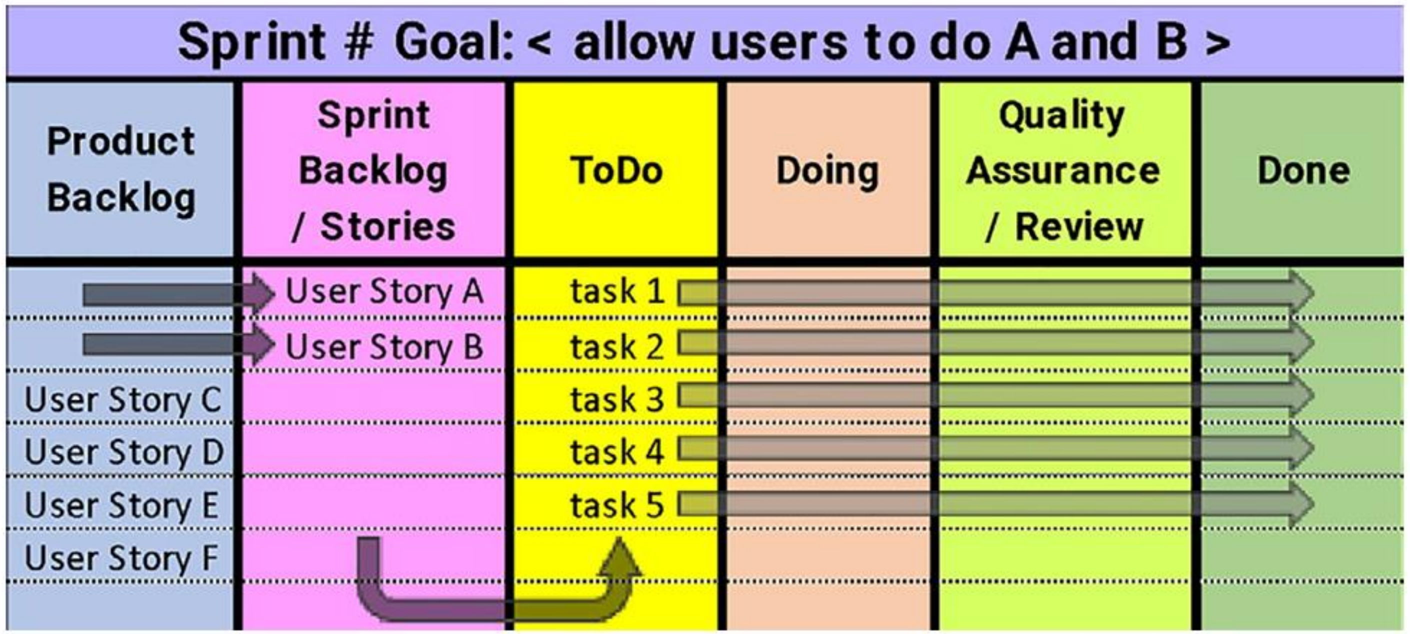




Scrum Board

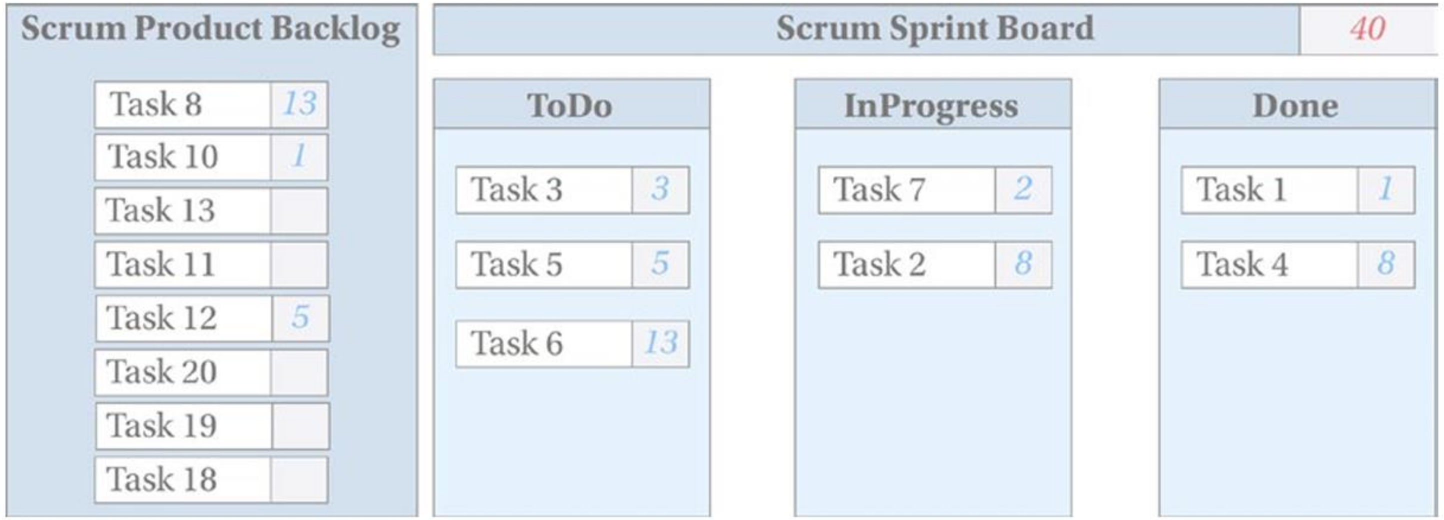
- The Scrum board may also include **Product Backlog** and **Sprint Backlog** columns, and additional workflow columns between **InProgress** and **Done** states such as
  - Review
  - Testing
  - Ready to Deploy
- Each team customizes the task stages according to its development process.

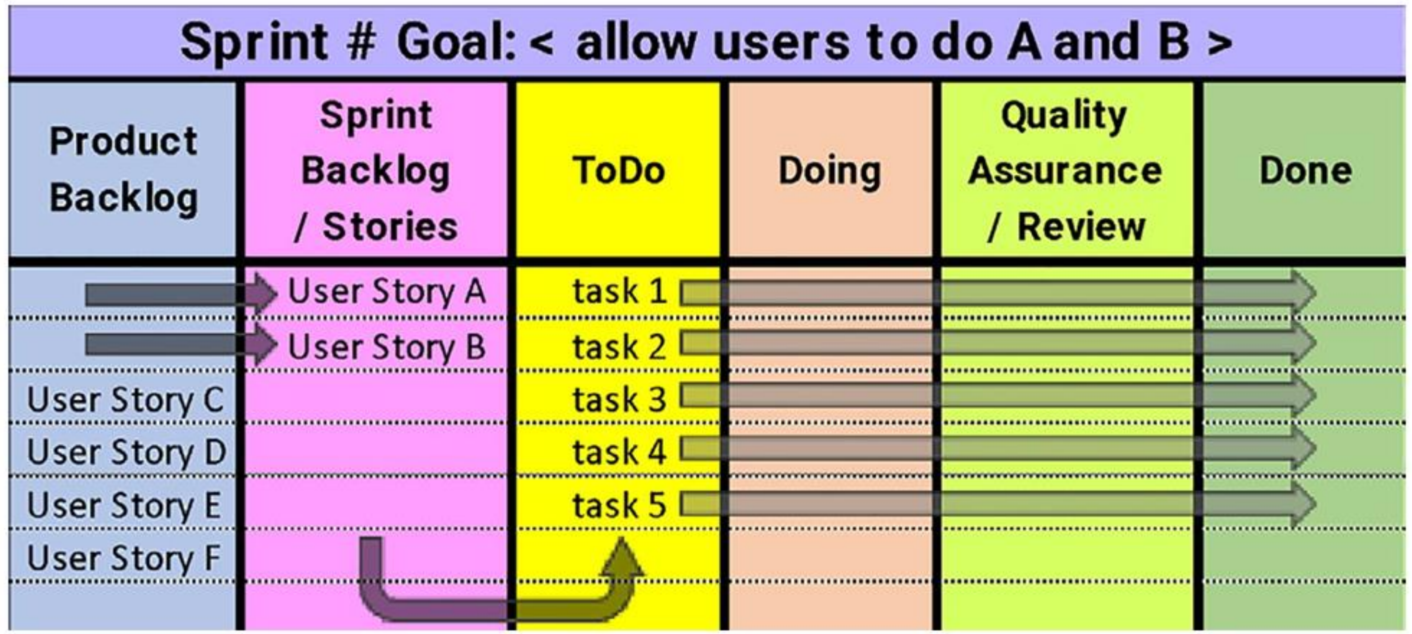




Scrum Board

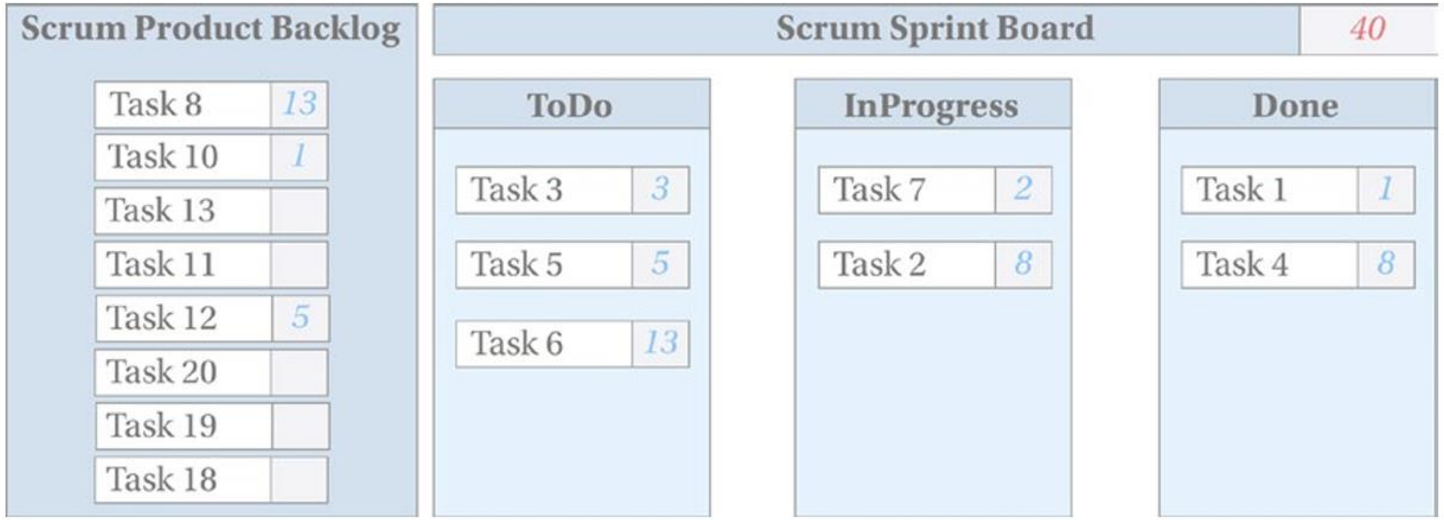
- Product Backlog contains prioritized items, some of them are estimated during backlog refinement meetings.
- During the sprint planning, the team selects some of the highest-priority items from the Product Backlog and moves them to the Sprint Backlog.
- These items are then decomposed into smaller tasks, which become the ToDo items for the sprint.

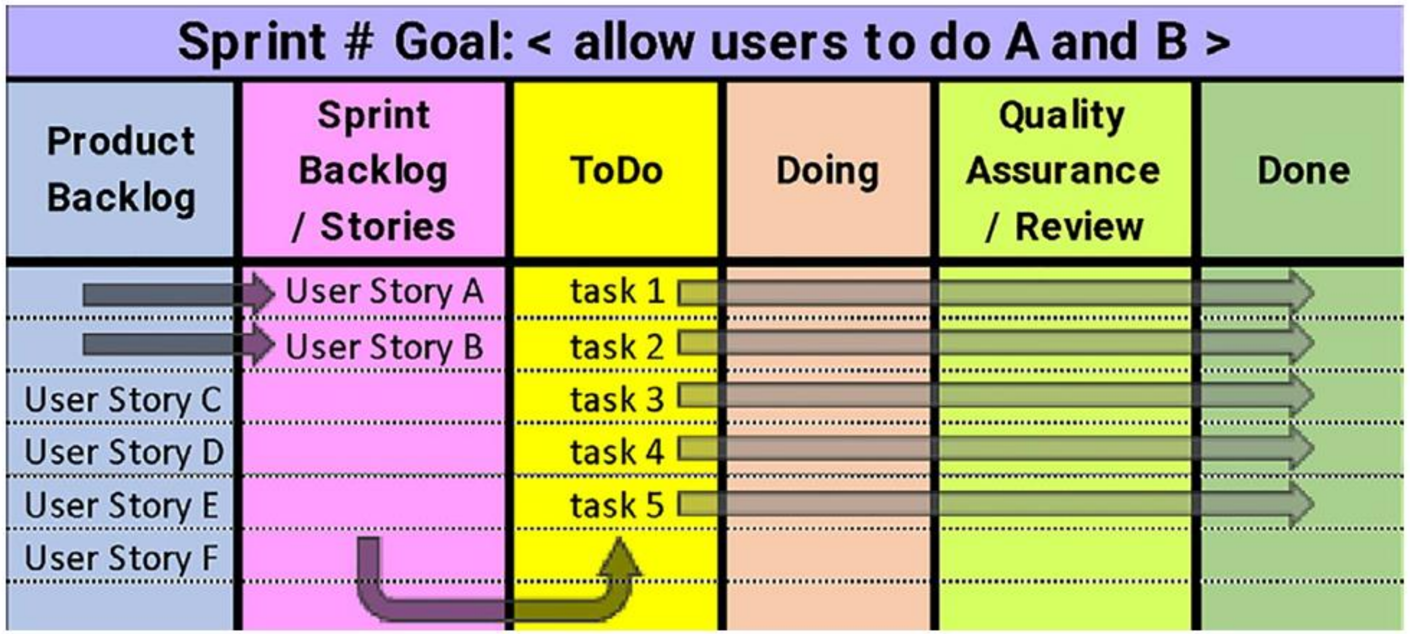




Scrum Board

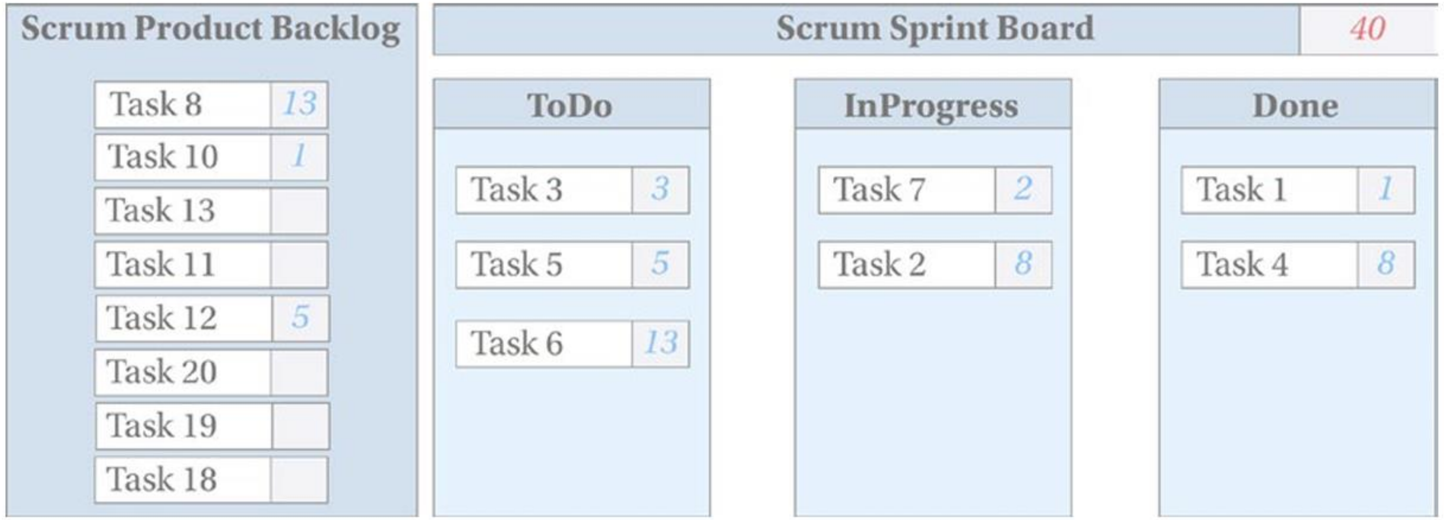
- These tasks move across the board until reaching the Done column once fully developed, tested, and integrated into the code base.
- The amount of work selected for the sprint is based on the team's capacity and historical velocity.
- The sum of estimations of the sprint tasks cannot exceed the team's average velocity.

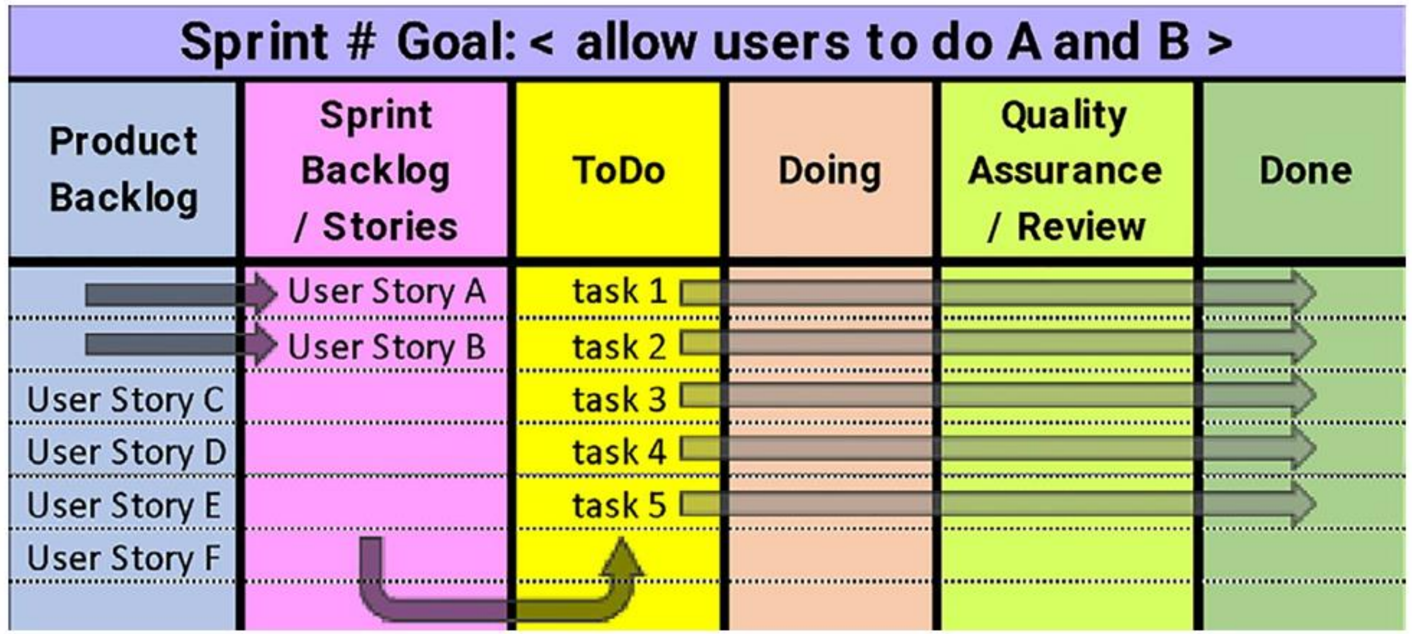




Scrum Board

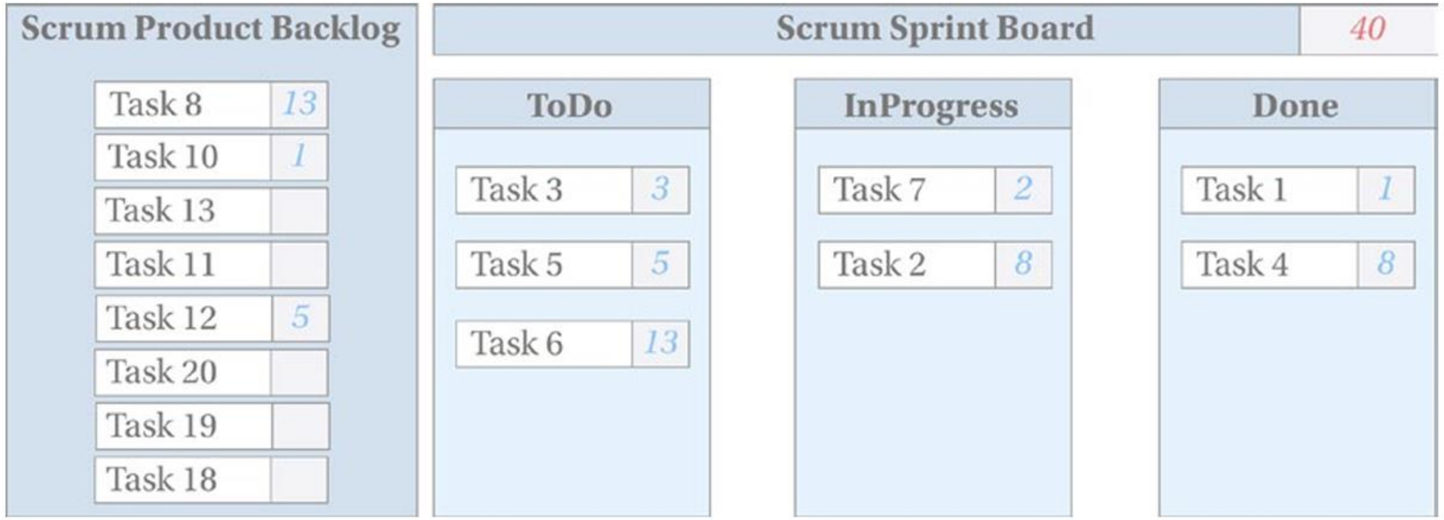
- This ensures that the team commits only to work that can be completed within the sprint.
- No one can add tasks to the sprint board during the sprint because it will affect the velocity in the end.
- Sometimes there are some exceptions, and the team can agree on moving the tasks to the sprint if some other task of the same estimation is removed from it.



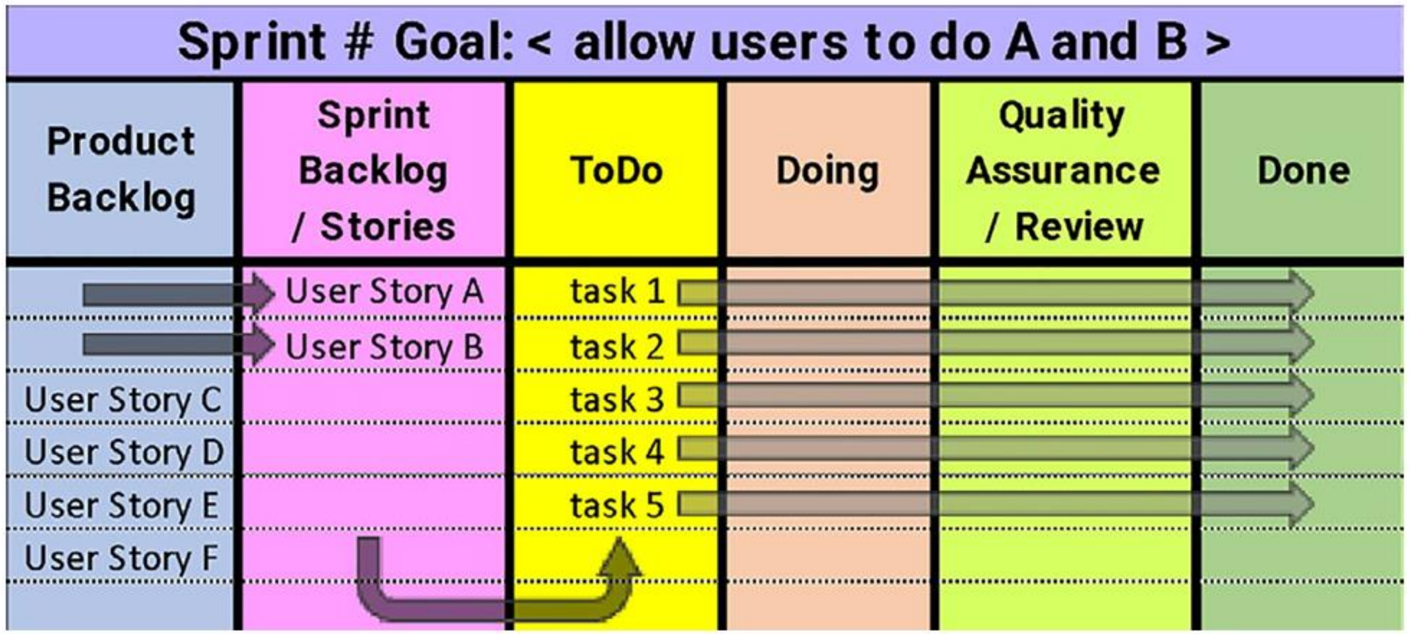


Scrum Board

- The figure above shows how the Scrum sprint board might look if ***the velocity of the team is 40 story points.***



- The figure below shows a generic single-sprint Scrum board. **Dark arrows** indicate the movement of items during sprint planning, while **light arrows** indicate the movement of items during the sprint.



Scrum Velocity

- The measure of the amount of work completed in a single sprint is called **velocity**. It helps the team decide how many Product Backlog items to include in each sprint.
- With most Scrum teams, ***estimation accuracy improves as the project progresses*** because the team gains experience and historical data from previous sprints. This effect in Scrum is called "**acceleration**".
- Over time, the ***productivity and velocity of the team may increase*** as the project progresses, because the team becomes better at breaking down and assigning tasks, creating more accurate estimates of the work needed for each task, as well as at collaborating with each other.

Scrum End of Project

- **After the last scheduled development sprint,** a final sprint may be done to bring closure to the project.
- This ***sprint implements no new functionality***, but ***prepares the final deliverable for product release***.
- It ***fixes any existing bugs***, ***finishes documentation***, and generally ***productizes the code***.
- Any requirements left in the product backlog are transferred to the next release.


Thank you